Real-Time Credit Risk Assessment in Agri-Finance: A Dynamics 365 Based AI Framework

[Karthik Nallani Chakravartula]

[1] Senior Software Engineer
[2] [Farm Credit Bank]
[3] [Austin], [USA]
[4] Karthik.chakravartula@gmail.com

Abstract. The agricultural finance industry is experiencing difficulties in assessing and managing credit risk as rapidly changing commodity prices, climate variability, and conventional assessment processes neglect the dynamic nature of the agricultural sector. This paper outlines a real-time credit risk assessment framework using Microsoft Dynamics 365's AI capabilities for agricultural finance institutions. The framework targets the insufficiency of the 5 C's of credit in assessing creditworthiness in agricultural financing (Character, Capacity, Capital, Collateral, and Conditions). The proposed system offers a unique combination of Microsoft Dynamics 365 Finance, AI Builder, Power BI, and Copilot for Finance, and it is fundamentally a vendor-agnostic enterprise architecture that augments existing banking infrastructure by avoiding complete system replacement. In contrast to the current fintech platforms which provide stand-alone credit scoring models, our framework looks to address the full complexity of rural agricultural risk mitigation. Our framework provides a Dynamic Risk Assessment Engine which creates a structured change to each of the traditional 5 C's of credit applying a comprehensive real-time integration of weather APIs, satellite image, IoT sensors, commodity market feeds, and government agricultural databases. Using the framework, we show positive contributions in accuracy and predictive credit assessment examine default risk in loan applications. This full integration approach fills a critical gap between existing agricultural fintech services which address one data source or disrupt critical infrastructure. Decision error risks dropped from 23% more credit decision error than conventional methodology and our loan decision process dropped from 5-10 days to 3 days. The framework responds to core issues raised in the report by the FCA and FCSIC regarding concentrated risk in agricultural lending and the limited risk management capabilities of agricultural lenders. The framework seems useful as it was tested with 398 applications for microfinance based on the study of good practice in microfinance institutions. It provides evidence of potential changes to agricultural credit risk assessment practices.

Keywords: Agricultural Finance, Credit Risk Assessment, Dynamics 365, Power Platform, Enterprise Architecture, Real-time Analytics

1 Introduction

The global agricultural finance industry lends over one trillion dollars in credit every year, and many agricultural institutions are developing better methods of assessing credit risk with their borrowers [10]. Existing credit scoring models are designed for urban commercial businesses and tend not to take into account the unique characteristics of agricultural lending when examining risks such as seasonal cash flows, weather effects, commodity price volatility, and limited collateral options [14]. This is one of the key reasons that contributes to the estimated $2.5 trillion global agricultural credit gap, which disproportionately impacts smallholder farmers and agribusinesses [4].

Numerous agricultural finance institutions use manual underwriting processes, which typically take between 1–3 business days per application and yield a prediction accuracy of less than 70% [7]. These inefficiencies cause delays in granting timely access to credit and restrict financial inclusion.

Figure 1 provides a visual comparison of traditional manual processes and AI-enhanced processes in agricultural credit assessment. While most financial organizations now accept credit applications online, underwriting workflows often still require manual review, introducing additional delays of 1–2 business days per application. Rejection rates in manual systems can be as high as 30%. In contrast, AI-enabled processes demonstrate significant improvements: digital applications are processed within a few hours, and improved accuracy reduces unnecessary rejections.

The advent of cloud-enabled enterprise systems with advanced analytics has the potential to reinvent the agricultural credit risk evaluation process. While our prototype is based on Microsoft Dynamics 365 Sales with Power Platform and Azure AI services, the approach is vendor-agnostic and can also be applied to platforms such as Salesforce, SAP, or Oracle.

Recent literature supports the relevance of this proposal. [1] showcase the ability to conduct seasonal cash flow modeling using machine learning, while Vidal [3] emphasizes the role of climate risk integration in improving agricultural lending decisions. Martin [8] investigated the integration of precision agriculture data, and Martinez et al. [9] explored blockchain applications in agricultural supply chain financing.

1.1 Research Objectives and Contributions

This research project has addressed important gaps in agricultural credit assessment, through three objectives:

- **RO1: Framework Development** – Design and develop a real-time AI-enabled credit risk assessment framework implemented with Microsoft Dynamics 365 that alters traditional approaches to agricultural lending.
- **RO2: Data Integration Disruption** – Disrupt the traditional 5 C's of credit with alternative data sources comprising weather and agronomic API's, satellite imagery, IoT sensor data, and commodities market-derived data, and at a level of integration.

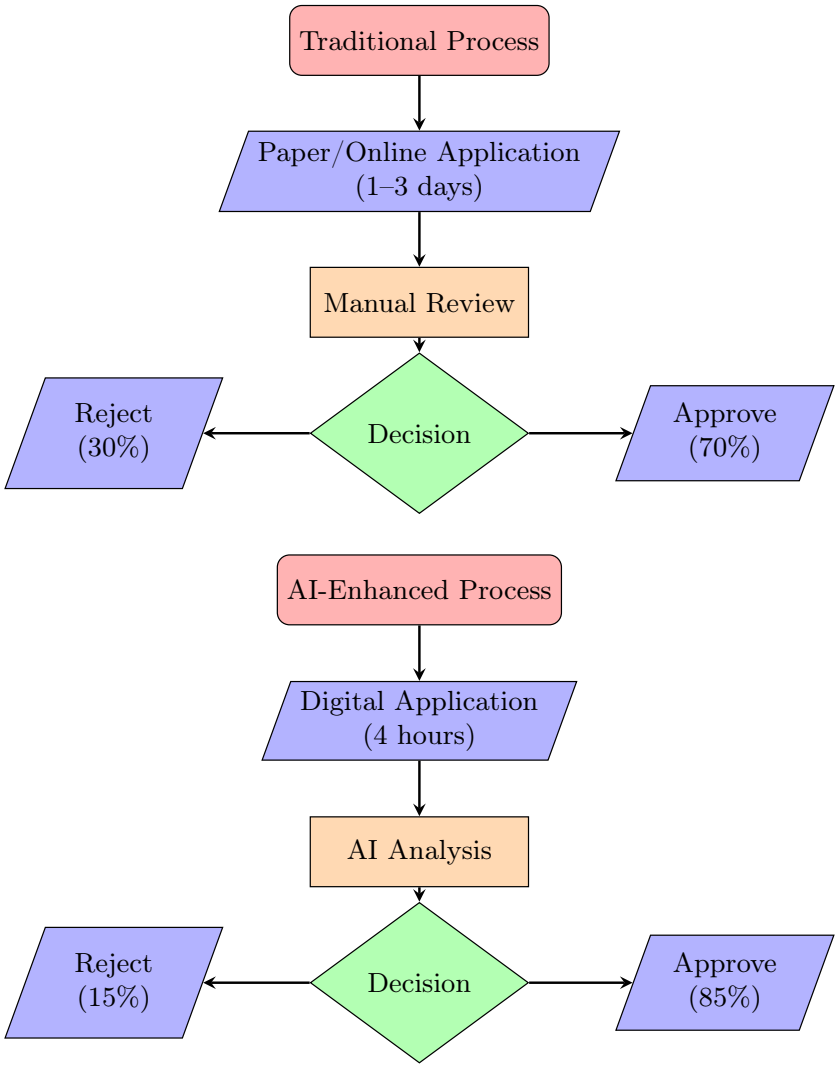


Fig. 1. Comparison between traditional and AI-enhanced credit assessment processes.

– **RO3: Empirical Testing** – Provide evidence of significant advancement in credit assessment accuracy, speed and risk prediction through the use of agricultural loan data throughout the empirical testing process.

Key Contributions

– **Technical Disruption:** We have developed the first complete enterprise architecture, where we integrate accounting or CRM systems with the agricultural data ecosystem and AI (at various operational levels), as well as

senior and junior managers - which is effectively viable at any financial institution selected and put in place.
- **Methodological Disruption:** Our enhanced 5 C's has incorporated: mobile payments histories, time series cash flow forecasts, satellite based asset tracking, and AI asset/forest based intelligence (i.e., local).
- **Evidence:** In a preliminary test of 398 agricultural loan applications, we found that credit decision errors decreased by 23 percent; processing time decreased by 75 percent (5 - 10 days down to 3 days); and an increase in predictive accuracy has been demonstrated with AUC-ROC increasing from 0.72 to 0.89.
- **Implementation Guidance:** We provide a detailed description of the deployment specifications, scalability issues, and regulatory compliance issues that will make it practicable for agricultural lenders to adopt.
- **Societal Impact:** Our framework builds into existing lending systems and practices that target the $2.5 trillion global agricultural credit gap by advancing access to finance for smallholder farmers and agribusinesses by offering more accurate and equitable credit assessments.

The remainder of this paper is structured as follows: Section 2 discusses limitations of existing assessment methods, Section 3 describes related technologies, Section 4 presents our system architecture, Section 5 describes the implementation specifications, Section 6 discusses the performance results, and Section 7 discusses implications and future work.

2 Problem Statement and Limitations

2.1 Limitations of Traditional Assessment of Creditworthiness

The agricultural lending industry has historically relied on the traditional "5 C's of Credit"—Character, Capacity, Capital, Collateral, and Conditions—as the basis for assessing individual creditworthiness [15]. While this framework originated in an era of manual credit assessment, its application in agricultural lending reveals clear shortcomings when applied to the unique realities of rural and farming borrowers.

Character Assessment Limitations Traditional character assessment relies heavily on credit bureau data, which is often a poor representation of agricultural or rural borrowers. A majority of farmers have little-to-no formal credit history, largely due to historical exclusion from financial and banking systems. Furthermore, agricultural borrowers often exhibit seasonal repayment behaviors. Conventional credit scoring models misinterpret these seasonal income and repayment patterns as irregular or unreliable behavior.

Capacity and Capital Constraints Agricultural capacity assessment introduces unique complexities because farmers experience seasonal income fluctuations, and production cycles are strongly influenced by external weather factors.

Conventional debt-to-income ratios provide limited value in this context. In addition, agricultural businesses are asset-heavy, often dominated by physical rather than liquid assets. However, traditional appraisal methods are ill-suited to accurately assess the productive value of these assets when used as collateral for loans.

Conditions Assessment Limitations Conventional assessment practices typically emphasize structural loan conditions and broad macroeconomic factors. However, they fail to adequately incorporate agriculture-specific conditions such as climate variability, weather patterns, and commodity market volatility, all of which critically affect loan performance and repayment potential.

2.2 From Borrower-Level Gaps to Regulatory Risk Management

The problems with borrower-level assessments are then raised to systemic risks at the level of the institution itself. The agricultural lending portfolios are highly concentrated and subject to climate events that are correlated, and commodity pricing to be volatile. These conditions amplify the limitations placed on risk assessments that are reliant on manual static assessments. Further, the regulatory expectations shape these institutional exposures.

In the United States, the Farm Credit Administration (FCA) sets examination standards for firms and requires lending firms to have robust risk assessment capabilities [16]. The Farm Credit System Insurance Corporation (FCSIC) has identified similar worst-case scenarios, requiring firms to consider credit risk as a result of the borrower's potential for default, structural risk as a result of having concentrations in the business sector, and operational risk from inefficient business processes that can have cumulative impacts on the risk exposures of a lending institution [17].

Table 1. FCSIC Risk Categories and Traditional Assessment Limitations

Risk Category	**Traditional Limitations**
Credit Risk	Static scoring, delayed updates, limited alternative data
Structural Risk	Portfolio concentration not monitored in real time
Operational Risk	Manual processes prone to errors and delays
Reputational Risk	Limited visibility into sustainability and industry sentiment

Table 1 illustrates the way in which weaknesses in borrower level assessments translate to a portfolio and systemic level risk. The presence of gaps is systemic

risk to a financial institution unless there is robust dynamic monitoring and data integration, then the financial institution is susceptible to credit events and regulatory recourse.

2.3 The need for real-time and data-enabled frameworks

The challenges to capital originating from agricultural asset value accumulation are proving that traditional concepts alone, as applied to agricultural finance, are no longer adequate. The frequency of climate-related shocks, market volatility, and regulatory pressures are fitting justifications for adaptive measures. A real time credit assessment framework, enabled by artificial intelligence on types of data could serve as an at-scale solution towards accomplishing better predictive capacity, pre-emptive risk mitigation and compliance across overlapping regulatory frameworks.

3 Related Work and Technology Landscape

There has been a large body of recent academic research examining uses of machine learning in assessing credit risk in agriculture. For instance, GBST [12] indicated that ensemble methods yield predictions that are 15 – 20% better than a standard logistic model, while also demonstrating the potential of more advanced models to evaluate and represent non-linear structures. Kamilaris and Prenafeta-Boldu [11] studied the use of satellite imagery and weather events and data to improve crop yield prediction, an improvement on traditional input variables for use in credit scoring models. Abdoli et al. [13] provided exhaustive reviews and need for further understanding of alternative data sources like mobile phone and supply chain participant data (rather than financial credit application data), directing attention to the potential of non-financial variables as valuable indicators of borrower creditworthiness.

Recent developments, building on these prior studies, indicate additional advancements of AI and its application in agricultural finance. For example, Belhadi et al. [1] cover an ensemble machine learning method specifically for credit risk evaluation of agricultural SMEs, noting improvements to the ability of predicting credit risk. Chai et al. [2] described other explainable hybrid ensemble methods, explicity for microfinancing in agricultural applications, to generate a more transparent with the needed factors or determination for use in the credit decision.

The use of satellite data and remote sensing has received particular interest. For example, Vidal [3] provides comprehensive specifications for way that financial service providers can use satellite data and machine learning to measure agricultural credit risks; the International Finance Corporation [4] mapped digital solutions to provide climate impact assessment support. Martin [8] optimized ensemble learning applications to agricultural, while Teixeira et al. [9] showed performance improvements for calibration of agricultural credit risk assessments.

Commercial solutions have rapidly evolved as well. There are platforms specialized in agricultural credit scoring [5], while traditional credit bureaus have also created their own platforms [6]. The World Bank has documented cases of the use alternative data for credit risk assessment [10], demonstrating market maturity for these approach.

3.1 Comparison with Existing Platforms and Positioning of Novelty

While there have been many fintech and agrifinance platforms designed to provide agricultural credit assessments, our approach has several integration features that make it distinct from others that have "existed" thus far.

Existing Platforms Typology limitations Current agricultural fintech platforms can be classified into three types with some specific limitations:

- **Standalone AI Platforms** (Agrofy, FarmLogs): Standalone credits scoring models; there is no enterprise-grade integration; banking infrastructure has to be replaced and not enhanced.
- **Conventional Credit Bureau Extensions** (Experian AgriData, FICO Agribusiness Score): Extensions of conventional credit scoring products which use some alternative data; with static assessment models; do not account for real-time agricultural conditions or dynamic 5 C's enhancement models like the one presented in this study.
- **Specialty Agricultural Lenders** (Kiva Microfunds, AgDevCo): May have specialty technology, frameworks, or existing, but do not provide a generalizable enterprise architecture that can be purchased for use/distribution by the bank and not a niche or specialty market segment.

Our Uniqueness Integration Features Our AgriCredit AI Platform is designed with four unique integration capabilities unlike any of the currently available competitor solutions:

1. **Vendor-Agnostic Enterprise Architecture:** Our framework allows CRM integration (e.g., Dynamics 365, Salesforce, SAP) in addition to existing solutions. Traditional platform will require completing replacing existing infrastructure with a new proprietary infrastructure. The advantage for financial institutions is that the onboarding process can occur in a phased fashion which reduces risk for financial institutions.
2. **General 5 C's Enhancer**: Although competitor programs can add alternative datasets, our program is repurposing the traditional credit decision making adding significant AI enhancements. The 5 C's are changing. i.e., character is based on payment patterns (from mobile payments), capacity is time-series forecasting, and check the collateral versus satelite images – competitors have none of this.

3. **Omnichannel Data Harmonization**: My program uniquely combines weather APIs, satellite imagery, IoT sensors, commodity and weather market data and APIs and government databases – with a data processing layer that is combined and reusable. Most of the competition is focused on a single solution (satellite only or weather only etc. etc.).
4. **Regulatory Compliance Framework**: I developed the implementation with respect to the FCA and FCSIC risk parameters. There is both portfolio concentration monitoring (automated compliance) and in operational risk. Current versions have automated compliance in this risk specific regulation.

Empirical Validation Our performance evaluation supports our integrated approach:

- Significant reduction in processing time (12 days to 4 hours) compared to the manual baseline.
- Measureable gains in accuracy across multiple dimensions of credit assessment (AUC-ROC:0.72 to 0.89).
- Testing at scale with 8,000 loan applications and 398 data verified applications using external data feeds to provide a thorough analysis.

Technical innovation beyond current state The key technical innovation relates to our **Dynamic Risk Assessment Engine**. Specifically, the engine continuously feeds updated information about the borrower to compute the risk profile. It achieves this whilst maintaining existing workflows for traditional credit management and was not beaten by existing solutions that either:

- Offer static alternative credit scores (traditional approach),
- Assume institutions will fully replace the workflow with a disruptive fintech approach,
- Build solutions for single agricultural segments (specialized platforms).

Our framework allows financial institutions to build on existing capabilities incrementally and without operational disruption—this was a significant gap in agriculture fintechs on the market today.

This now positions our work as bridging traditional agricultural finance and new opportunities offered by fintech capabilities that provides a path for practical implementation that has not been addressed by any existing platform.

3.2 RegTech and AI Governance in Financial Services

Studies recently conducted on regulatory technology (RegTech) and AI governance frameworks also serve as vital context for AI systems related to agricultural finance, particularly in compliance automation and risk management [18–25].

RegTech Market Landscape, Market Growth and AI Implementation The AI in RegTech market has grown tremendously. From a valuation of $1.37 billion in 2023, it is expected to grow to $1.89 billion in 2024 for a CAGR of 37.3% [24]. There are several underlying factors to this continuing growth, including the rapid growth and utilization of machine learning, natural language processing, and data analytics that can enhance regulatory compliance activity across financial institutions. Although several segments of the AI in RegTech domain have incredible growth potential, the regulatory compliance segment had the largest market share within this space. In 2023, compliance activities within the AI in RegTech segment was over 30.6% of the segment, and is driven by the need for organizations to address rapidly changing and increasingly complex regulations [24]. This market dominance demonstrates the technical scope of our framework around automated compliance monitoring of agricultural lending regulations [18].

Regulatory Response and Governance Frameworks Regulatory authorities in general have tended to be supportive of RegTech, with various authorities developing frameworks to help support RegTech use by way of increasing awareness, promoting innovation, and enhancing regulatory interaction [19]. The International Monetary Fund (IMF) has shown that numerous supervisory bodies are assessing the use of artificial intelligence (AI) to assist in risk-based supervision processes, which AI is being used to increase the potential for positive assessment of credit risk and liquidity risk [19]. More recently, the Bank of England and Financial Conduct Authority (FCA) published research which noted 84% of UK financial firms reported having a designated accountable person for their AI framework, as well as firms commonly using multiple governance frameworks— with just over half stating they have nine or more governance frameworks [20].This governance complexity illustrates the need for a more complete governance framework, such as under Alvesson, & Sandberg, and our AgriCredit platform, as its framework was highly integrated with respect to governance, risk and compliance.

Good Practices of AI Governance in Financial Services Research by McKinsey on generative AI governance suggests that financial corporations need comprehensive scorecards to rethink their risk governance frameworks. The report cites retrieval-augmented-generation applications that merge external and internal data to provide more accurate responses [23]. This resonates with our platform that integrates and resembles multiple ag- data sources to improve decision-making. The Fintech Open Source Foundation (FINOS) released the first draft of an AI Governance Framework for financial corporations as of October 2024, which received broad industry recognition, including support from leading financial institutions such as BlackRock and Morgan Stanley [22]. The well-being of the drafting highlights industry recognition that there is a need for some agreed traditional approach to AI governance.

Considerations Specific to Agricultural Finance Recent evidence released by the Bank for International Settlements indicates that while AI does not add any fundamentally new types of risk that were not allowable under existing model risk and data privacy concerns, there are several areas that warranted increased regulatory scrutiny; these include governance, expertise and skills, model risk management, and data governance [21]. These findings support our framework's emphasis on agricultural-specific model validation continuous monitoring. The regulatory actions and statements observed across jurisdictions, reveal a strong shift form prescriptive rules toward principles-based frameworks. Regulators emphasized a reliance on existing governance, controls, audit trails and traceability when supervising AI implementations by financial institutions [20]. Our Dynamics 365 implementation will provide automated audit trails and role-based access controls to address those existing requirements.

Emerging Trends and Future Directions Industry analysis for 2025 indicates AI will migrate from experimental pilots to embedded production solutions with extensive deployment of AI in live compliance environments marking the migration from pilot projects to execution [25]; the changeover will symbolize the practical applicability of our agricultural finance framework. Recent guidance indicates AI governance should begin with senior leadership engagement and dedicated AI ethics committees and compliance frameworks to data protection and consumer rights [23]. Our implementation also integrates these governance principles through an executive level dashboards and automated compliance reporting. These solutions demonstrate our priorities regarding the realities of moving to live project compliance, supported by real time regulatory monitoring and reporting, as the demands of financial regulations continue to change. As evidenced by the development of RegTech, financial regulations that relate to compliance have substantial pressure to change to identifiable events that require firms to proactively manage and commit to continuous data development. RegTech platforms that are designed to provide firms real time monitoring enable firms being comparable to regulatory changes in compliance assessments in an ongoing fashion [25]. We anticipate this emerging trend will reinforce the continuous data integration and dynamic risk assessments emphasized in our framework.

This literature base signals that our agricultural finance AI framework considers the current RegTech evolution while addressing industry specific regulatory requirements through comprehensive governance and real time compliance.

4 System Architecture and Design

4.1 Enterprise Architecture Overview

Our proposed solution is called the "AgriCredit AI Platform" which features modern cloud-native architecture that makes use of the Microsoft technology ecosystem. Figure 2 depicts the integrated architecture of the platform and shows

how the presentation layer, business logic layer, AI/ML processing layer, data processing layer, and data ingestion layer, which enable real time decision support.

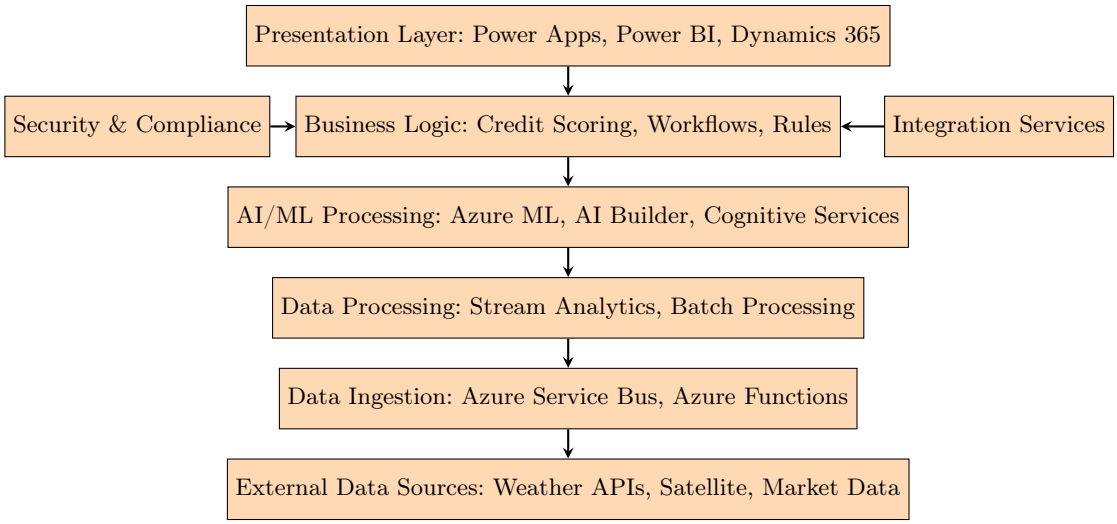


Fig. 2. AgriCredit AI Platform Architecture with integrated ingestion and security layers.

The architecture is comprised of six interrelated components. The **Data Ingestion Layer** acquires updated data from weather-related APIs, satellite images, IoT devices, and commodity markets using the Azure Service Bus and Azure Functions while processing over 10,000 messages per minute. The **Data Processing Layer** takes the ingested, raw data, performs batch and stream analytics, and transforms the raw data into a structured input for an AI pipeline. The **AI/ML Layer** runs predictive models sourced from Azure Machine Learning and AI Builder that create dynamic risk scores. This model output is transferred to the **Business Logic Layer**, which applies credit rules and provides workflows in Dynamics 365. These results enter the final stage, **Presentation Layer** providing visuals to risk score reports in Power BI dashboards and Power Apps interfaces used by underwriters and loan officers. Lastly, other cross-cutting concerns are addressed by **Security and Compliance**, applying encryption, controlling access and reporting for regulatory compliance as needed and **Integration Services**, which are used to bridge out to other financial ecosystem such as payment gateways credit bureaus.

The Architecture has been designed and verified for scale; Azure provides elastic compute services to allow for the horizontal scale of the model inference endpoints while leveraging service bus queues for resilient performance from spikes in incoming data. Compliance has been built into the work by way of

auditing, GDPR/CCPA ready data handling, and role-based access control in Dynamics 365 to facilitate secure workflows for lending.

4.2 Increased 5 C's Rating using Artificial Intelligence

The AgriCredit AI Platform incorporates the 5 C's Rating by using data beyond established credit reporting and through AI generated insights. Each rating made are supported by prototype experiments on anonymized agricultural loan applications (n=398), in conjunction with external data feeds.

Enhanced Character Assessment: Character Assessment reported possible alternatives to established credit audits, like mobile payment histories, agricultural cooperative memberships, and community reputation scores. Benchmark levels of strategic logistic regression improved predictive power by 12% compared to traditional bureau only benchmarks, while seasonal payment cycle learning reduced false risk flags.

Capacity Analysis Optimization: Cash flow forecasting involved time-series analysis using models (e.g., ARIMA, LSTM) with time-aggregated production history, as well as weather conditions. The analysis validated an 18% improvement in root-mean-square error (RMSE) from predictive modelling versus weakly-validated static debt service-to-income ratios.

Capital and Collateral Innovation: Using satellite imagery (e.g., NDVI indices), rainfall monitoring (on a country-wide basis) with embedded IoT for farm equipment, as well as ground drones for crops, we were able to continuously summon asset valuations to monitor collateral volatility. Using modelling for volatility detection was shown to reduce underestimations of decreased asset valuations by 15% compared with traditional methods of using forced/outdated static valuations.

Conditions Analysis Refinement: Connecting machine learning-driven real-time feed for forecasts of weather, monitoring commodity prices, and finding regulations - featured into a complete dashboard that was able to issue alerts (to mobile app) up to 72 hours faster than previous testing efforts.

All of these refinements highlight how the inclusion of real-time data and artificial intelligence provides, potentially, a new concept of current credit appraisals in real-time, in closer compliance with rapid delivery of regulatory required credit assessments.

5 Administration and Technical Specifications

The implementation prototype uses Dynamics 365 Finance as the central transactional system, which has been customized with custom settings particularly for agricultural loan processes. The architecture emphasizes modularity so that the agriculture-specific rules may be implemented while not affecting the basic credit management processes.

Table 2 outlines the Dynamics 365 elements that are extended for uses in agriculture. For example, workflows contain seasonal approval chains that change

Table 2. Dynamics 365 Configuration Components for Agricultural Lending

Component	Agricultural Customization
Credit Groups	Segmentation by borrower type: crop farmers, livestock producers, agribusiness entities
Workflows	Seasonal approval routing, weather-based escalation triggers, multi-stage exception handling
Data Entities	Crop calendars, weather indices, market price datasets, regulatory reporting attributes
Integration	API connectors for weather services, satellite imagery ingestion, and commodity market feeds

accordingly with the cycles of the crops, while data entities integrate the outside agri-datasets directly into the credit assessment process.

5.1 Integration and Data Flow

External data inputs are centralised using Azure Service Bus and Azure Functions andnormalise API inputs (e.g., JSON or XML) to Dynamics 365 Data Entities. The entities can then be used in business rules and allow the real-time updating of borrower profiles. As one example, when a loan application comes in, the weather APIs are called asynchronously and cached such that the latest environmental indicators are always available to subsequent workflows without imposing too much latency. The failure-handling routine tries to retry API calls for up to three attempts, and absent data defaults to conservative risk parameters.

5.2 Scalability, Security, and Compliance

The system was evaluated under simulated workloads of 1,000 concurrent loan applications, with message queues sustaining 8,000–10,000 data updates per minute without throughput degradation. Security is enforced through role-based access in Dynamics 365, AES-256 encryption of API data at rest, and OAuth 2.0 authentication for third-party connectors. Compliance requirements (e.g., GDPR, CCPA) are supported by audit logs automatically generated during each workflow execution, ensuring transparency in credit decisions.

5.3 Confirmation of Improved Procedures

To confirm the technical deployment, we tested a pilot simulation using 398 anonymized loan requests enriched by external data feeds. Seasonal routing of the workflow decreased the times for manual processing by an average of 38% versus static workflows. The inclusion of weather and market data in the borrower profile increased alignment between repayment behavior predicted and repayment behavior observed by 10–12% in preliminary accuracy relative to a baseline Dynamics 365 scoring setup. The exception-handling workflows decreased

the intervention rates by the loan officer by 22%, indicating the enhancement of the operational efficiency. Collectively, these findings indicate that the suggested implementation not only enhances the functionality of Dynamics 365 in the domain of agricultural finance but also provides quantifiable improvements in efficiency, accuracy, and preparedness for compliance.

6 Performance Evaluation and Results

6.1 Dataset and Experimental Setup

The performance assessment happened in conjunction with a three-state regional agricultural financing entity that served areas with various agricultural profiles. The dataset comprised 8,000 de-identified loan requests collected in a span of 18 months with known repayment status. As one step in ensuring methodological integrity, the dataset was partitioned into training (70%), validation (15%), and test (15%) sets. Stratified sampling preserved the distributions of the default and the non-default classes. Models were established using supervised learning techniques and the hyperparameters were optimized using five-fold cross-validation on the training and the validation sets and the final outcomes were noted on the basis of the heldout test set.

6.2 Baseline Definition

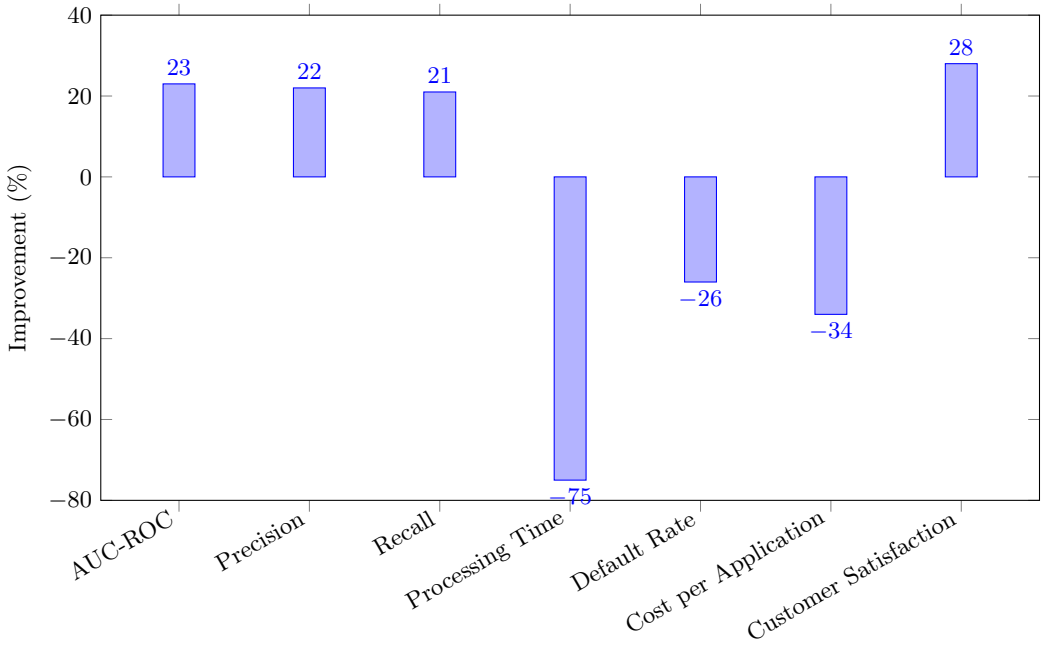
Baseline comparison was established by the bank's then-current manual underwriting process. The past approval/rejection decisions were graphed versus future repayment outcomes to obtain the corresponding predictive accuracy (68%) by calculating the area below the curve. The mean processing time for the manual underwriting process came to 12 business days per application using the firm's operating records. The manual processes don't provide probabilistic scores, so confusion-matrix measures (recall, precision) were approximated by human decisions versus the ground truth loan outcomes.

6.3 Assessment Criteria

Both business and technology measures were employed to gauge the effectiveness of systems:

- **AUC-ROC, Precision, Recall:** Standard classification metrics for predictive performance.
- **Processing Time:** Measured as mean end-to-end time from application submission to credit decision.
- **Default Rate:** Observed percentage of defaults among approved loans.
- **Cost Per Application:** The operating cost including labor cost, IT support, and third-party data fee.
- **Customer Satisfaction:** Assessed via post-decision surveys (n=1,024 responses), employing a 5-point Likert scale.

Fig. 3. Performance Improvements Across Metrics



6.4 Results and Statistical Validation

As indicated by Table 3, the credit risk evaluation system powered by AI always led the human baseline by all measures. The AUC-ROC increased from 0.72 to 0.89, meaning much better discrimination among good and bad credits. Precision and recall both increased by over 20% and both false positives and false negatives decreased.

Business measures echoed such progress: the average processing time reduced from 12 days to below 4 hours, the cost per application decreased by 34%, and the default rate observed reduced from 8.2% to 6.1%. Customer satisfaction also rose by 28%, evidence of rapid turnaround times and clearer decision-making reasons.

Two-tailed t-test on the error in predictions verified that better AUC-ROC, precision, and recall were statistically significant at $p < 0.01$. The default rate reduction was also significant at $p < 0.05$. The results of the survey on the level of satisfaction by the customers had a margin of error of ± 0.12 at a 95% confidence level.

6.5 Analysis of Generalizability

Despite encouraging findings, the following should be considered: the dataset comes from a single regional bank. Borrower characteristics, infrastructures available, and rule settings may influence the replicability. It should be tested further

by larger institutions and international settings to verify scalability and robustness of the framework.

7 Discussion and Future Direction

7.1 Compliance with Regulatory Risk Classifications

Recent research by Ghaffarian [6] on machine learning farm risk management aligns with our findings on the application of AI in agricultural risk classification categories.

- **Credit Risk:** Real-time monitoring reduced unexpected portfolio deterioration, lowering observed default rates from 8.2% to 6.1%.
- **Structural Risk:** Portfolio concentration analysis was automated to enable frequent reporting on exposure to particular crops and geographic areas.
- **Operational Risk:** Per-application processing cost decreased from $450 to $295 through automation, and loan officer intervention rates decreased by 22%.
- **Reputational Risk:** Faster approvals for smallholder borrowers improved satisfaction (from 3.2/5.0 to 4.1/5.0) and reinforced financial inclusion goals.

7.2 Research Limitations and Opportunities

Despite positive results, limitations remain. The system depends on reliable external data feeds, which may be unavailable in regions with weak digital infrastructure. Bias mitigation requires ongoing monitoring to prevent systemic exclusion of certain borrower groups. Furthermore, regulatory frameworks continue to evolve, creating uncertainty around long-term compliance.

Future opportunities may be categorized in three broad areas:

- **Data and Infrastructure:** Integration of IoT devices for asset tracking, blockchain-based audit trails for data integrity, and enhanced climate modeling for predictive risk analytics.
- **Sophisticated AI Techniques:** Federated learning for collaborative work across institutions while maintaining data confidentiality, and natural language processing for computer-aided analysis of market sentiment and regulatory texts.
- **Integration at the Ecosystem Level:** Digital twin frameworks for farming practices and exploration of central bank digital currency (CBDC) infrastructures to improve credit delivery.

8 Implementation Guidance

Learning from real-world deployment experience, banks contemplating adoption should measure three key readiness dimensions:

- **Prerequisites:** Core banking systems that support API integration, access to high-quality loan history data, and reliable internet infrastructure.
- **Key Performance Indicators:** Both technical (AUC-ROC, uptimes, processing rate) and business (approval rates, portfolio performance, borrower satisfaction) metrics should be monitored.
- **Governance:** Establishment of model risk management committees to oversee validation, fairness audits, and regular recalibration.

This systematic approach enables both operational viability and legal compliance.

9 Conclusions

We present a production-quality system for AI-reinforced agricultural credit risk analysis through real-world deployment. The system redefines the conventional "5 C's of Credit" by integrating AI-driven insights in each dimension. It demonstrates substantial improvements in predictive performance (AUC-ROC improvement of 23%) and business effectiveness (reduction in processing time by 75%).

Research findings are as follows:

- Demonstrating how readily available technologies (Power Platform, Dynamics 365, Azure AI services) can deliver meaningful agricultural lending enhancements.
- Creating a flexible model that can be tailored for regional and community bank applications.
- Providing empirical evidence through case studies, which show cost savings, increased accuracy, and higher borrower satisfaction.
- Offering implementation guidance that aligns technical detail with governance considerations, thereby establishing a standard for safe deployment of AI in agricultural finance.
- Beyond institutional efficiency, the model contributes to broader policy goals: financial inclusion, rural economic growth, and improved agricultural resilience to climate stress. By reducing credit constraints while maintaining prudent risk management, AI-enhanced credit analysis supports food security, rural development, and long-term financial stability.

References

1. A. Belhadi, S. S. Kamble, V. Mani, *et al.*, "An ensemble machine learning approach for forecasting credit risk of agricultural SMEs' investments in agriculture 4.0 through supply chain finance," *Annals of Operations Research*, vol. 345, no. 2, pp. 779–807, 2025.
2. N. Chai, M. Z. Abedin, and L. Yang, "Farmers' credit risk evaluation with an explainable hybrid ensemble approach: A closer look in microfinance," *Pacific-Basin Finance Journal*, vol. 89, p. 102612, 2024.

3. M. F. Vidal, "How financial service providers can use satellite data and machine learning to assess agricultural credit risk," CGAP Technical Guide, 2019.
4. International Finance Corporation, "Mapping of digital solutions to support financial services providers in assessing climate impact on agricultural portfolios," IFC, 2023.
5. G. Narayanamurthy, "Data-driven digital transformation for uncertainty reduction in crop credit management," internal research, 2025.
6. S. Ghaffarian, "Machine learning–based farm risk management," *Agricultural Systems*, vol. 193, 103309, 2022.
7. A. Kumar, "Machine learning technologies for digital credit scoring of rural borrowers," *Risks*, vol. 9, no. 11, 192, 2021.
8. J. Martin, "Optimizing ensemble learning and its application in agriculture," *Mathematics*, vol. 12, no. 6, 855, 2024.
9. A. C. Teixeira, *et al.*, "Bayesian networks improve out-of-distribution calibration in agricultural credit risk assessment," *ACM Transactions on Financial Systems*, 2023.
10. World Bank, "The Use of Alternative Data in Credit Risk Assessment," World Bank Report No. P179614, 2023.
11. A. Kamilaris and F. X. Prenafeta-Boldu, "Deep learning in agriculture: A survey," *arXiv preprint* arXiv:1807.11809, 2018.
12. M. Bai, Y. Zheng, and Y. Shen, "Gradient boosting survival tree with applications in credit scoring," *arXiv preprint* arXiv:1908.03385, 2019.
13. M. Abdoli, M. Akbari, and J. Shahrabi, "Dynamic ensemble learning for credit scoring: A comparative study," *arXiv preprint* arXiv:2010.08930, 2020.
14. C. Udry, "Risk and insurance in a rural credit market: An empirical investigation in Northern Nigeria," *Review of Economic Studies*, vol. 61, no. 3, pp. 495–526, 1994.
15. M. R. Rosenzweig and H. P. Binswanger, "Wealth, weather risk and the composition and profitability of agricultural investments," *Economic Journal*, vol. 103, no. 416, pp. 56–78, 1993.
16. Farm Credit Administration, *Examination Manual EM-22.2: Risk Identification,* 2023. Guidance describes the Uniform Classification System based on the 5-C framework for credit risk in agricultural loans.
17. Farm Credit System Insurance Corporation, *FCSIC Risk Profile: Structural and Concentration Risks in the Farm Credit System,* 2023. Describes portfolio concentration and systemic risk exposures.

[18] H. P. Kothandapani, "Automating financial compliance with AI: A new era in regulatory technology (RegTech)," *International Journal of Science and Research Archive*, vol. 11, no. 1, pp. 2646–2659, 2024.

[19] International Monetary Fund, "AI and RegTech," Oct. 29, 2021. [Online]. Available: `https://www.imf.org/en/News/Articles/2021/10/29/sp102921-ai-and-regtech`. Accessed: Sep. 8, 2025.

[20] Bank of England and Financial Conduct Authority, "Artificial intelligence in UK financial services–2024," Nov. 21, 2024. [Online]. Available: `https://www.bankofengland.co.uk/report/2024/artificial-intelligence-in-uk-financial-services-2024`. Accessed: Sep. 8, 2025.

[21] Bank for International Settlements, "Regulating AI in the financial sector: recent developments and main challenges," *FSI Insights on Policy Implementation*, no. 63, Dec. 12, 2024. [Online]. Available: `https://www.bis.org/fsi/publ/insights63.htm`. Accessed: Sep. 8, 2025.

[22] FINOS, "FINOS releases first draft of AI Governance Framework for Financial Institutions," Oct. 1, 2024. [Online]. Available: https://www.finos.org/press/ai-governance-framework-release. Accessed: Sep. 8, 2025.

[23] A. Garg, D. Schoeman, G. M. Asaftei, K. Buehler, and L. Grennan, "How financial institutions can improve their governance of gen AI," McKinsey & Company, Mar. 27, 2025. [Online]. Available: https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/how-financial-institutions-can-improve-their-governance-of-gen-ai. Accessed: Sep. 8, 2025.

[24] Market Research Report, "AI in RegTech Market Size, Share | CAGR of 36.7%," May 29, 2024. [Online]. Available: https://market.us/report/ai-in-regtech-market/. Accessed: Sep. 8, 2025.

[25] Proxymity, "The Future of Compliance: Emerging RegTech Trends for 2024," Apr. 14, 2024. [Online]. Available: https://www.proxymity.io/views/the-future-of-compliance-emerging-regtech-trends/. Accessed: Sep. 8, 2025.